

OPTIMIZATION TECHNIQUES

Tutorial 1

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Topics: Mathematical Review & Unconstrained Optimization

Q1. (a) Using the definition of a convex set, prove that $S1 = \{x \in \mathbb{R}^2 : x_1 + 2x_2 \leq 4, x_1 \geq 0, x_2 \geq 0\}$ is convex.

(b) Show, with a counterexample, that $S2 = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \geq 1\}$ is NOT convex.

(c) Define an ε -neighbourhood of a point x_0 in $\mathbb{R}^n$. Determine whether $S1$ is a polyhedron, and if so, express it as an intersection of half-spaces (hyperplanes).

[BTL-3]

Q2. Find the equation of the hyperplane in $\mathbb{R}^3$ passing through the points $P1 = (1, 0, 1)$, $P2 = (0, 1, 1)$, and $P3 = (1, 1, 0)$. Express the hyperplane in the linear-variety form $\{x : a^T x = b\}$.

[BTL-3]

Q3. For $f(x, y) = x^2 + 2y^2$, find the gradient at the point $(1, 1)$ and the equation of the level curve through $(1, 1)$. Analyze and show that the gradient vector is perpendicular to the tangent of the level curve at that point.

[BTL-4]

Q4. Expand $f(x, y) = e^x \ln(1 + y)$ in a Taylor series about the point $(0, 0)$, retaining terms up to second order.

[BTL-3]

Q5. For $f(x, y) = x^3 + y^3 - 3xy$, find all critical points. Using the first-order necessary condition and the second-order (Hessian) sufficient condition, classify each critical point as a local minimizer, local maximizer, or saddle point.

[BTL-4]

Q6. (a) Using the golden section search method, perform two iterations to reduce the interval of uncertainty for minimizing $f(x) = x^2 + 2x$ over the initial interval $[-3, 2]$.

(b) Briefly explain how the Fibonacci search method's interval-reduction sequence differs from that of golden section search, and state one advantage of Fibonacci search over golden section search.

[BTL-3]

Q7. (a) Apply the bisection method to locate the minimizer of $f(x) = e^x - 2x$ by finding the root of $f'(x) = 0$ on $[0, 1]$, performing three iterations.

(b) Apply Newton's method to the same function starting at $x_0 = 0$, performing two iterations.

(c) Analyze and compare the convergence behaviour (rate, robustness, and assumptions) of the two methods.[BTL-4]

— End of Tutorial —